

Bingo Word Finders

eHealth Students Lab Project

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1. Introduction

As we all know, medical imaging is a crucial aspect of the healthcare field, and getting to know as many imaging modalities as possible is key for all of us, the MAIA and MIC students. During the development of the projects in which we have been involved this semester, we have been given the opportunity to learn how to work with some modalities and specific types of medical images. This includes brain MR (Magnetic Resonance) images, skin lesion images, lung CT (Computed Tomography) images, etc. However, there are still some organs and modalities that remain unexplored to us, like ultrasound, PET (Positron Emission Tomography), or even an MRI or CT in a different file format such as DICOM (Digital Imaging and Communication in Medicine). For this reason, our proposal for the Students Lab of the eHealth course consists of a game that allows students and participants to explore some of these unknown types of images.

Understanding as many modalities and medical images as possible is essential for us, being students involved in the field of medical imaging. Thus, within our class, we conducted an educational game designed to enhance our understanding of many different medical imaging modalities. The main objective of the proposed Bingo Word Finders game was to provide a hands-on and collaborative learning experience that would allow students to interact with various imaging modalities in a fun and interactive manner. This approach aimed to motivate the students to acquire some knowledge regarding topics that have not been explained before.

The key of learning through games is to motivate the participants to accomplish a goal, so our proposal consisted of finding the secret words, letter by letter. The game was based on a 3x3 Bingo board and was played through a website that provided the code and images to perform each task. Each group worked independently and collaboratively as a team to solve the different challenges regarding the different modalities and, once they finished three of the tasks (a line in the bingo board), they were given some letters that will help them uncover the secret words. The first team to guess the secret word became the winner.

This innovative approach to learning aimed to make the study of medical imaging more interactive and memorable. It also provided small pieces of code that the participants could further use when they faced similar challenges or tasks in the future. By engaging students in a dynamic and competitive environment, the game promoted a deeper understanding of the imaging modalities and formats presented. Additionally, it also helped to build teamwork and communication skills, which are crucial in any future profession.

In the following sections of this report, we will detail every step in the process of developing this activity for the Students Lab, including the methodology, the game itself, and the feedback received from the participants.

2. Brainstorming

The first step of this process was a brainstorming phase. This project has an element of creativity, so we decided to explore the ideas each group member could come up with and work from that base. Initially we decided on some key elements for our project.

- **Educational:** The end goal of the activity was to teach a concept to our classmates. For that, we had to analyze what we had learned during our studies and find a topic that we would have liked to learn more about.
- **Entertaining:** We wanted our proposed activity to be engaging and ensure all participants had fun. This would also help our fellow classmates remember the main topic of the student lab better.
- **Originality:** A fresh approach to learning while having a topic that stood out was also key, as it would make our lab more memorable and participants would have their interest piqued.

With these objectives set up, we took a week to individually come up with ideas. After that time period, we had a brainstorming session and discussed the following possible options:

- **Find Words:** This would be a game on which players would have to solve small challenges in order to obtain letters from a secret word that they have to guess. The focus of the challenges would be explore different image modalities, specially ones that we have not worked with during the semester. One great point of this proposal was that a competitive game is usually very engaging and will make people work harder in order to win.
- **Bingo:** The well known game of bingo can be adapted for our purpose. Each number could have a challenge associated that people would complete in order to finish their individual bingo cards. This game also has a factor of randomness, which makes the activity more exciting. The topic surrounding the challenges would have to be decided.
- **MRI reconstruction using Blender [1]:** During the semester we have worked a lot with MRI and we have learned how useful spacial information can be, so knowing how to obtain a 3D volume with these images could be interesting. However, this proposal would require every group of participants to have a laptop with the program Blender installed and it would not run optimally on every laptop.
- **Deep Learning model implementation:** Another approach more technical focused was to teach how deep learning models are implemented and apply such models to medical images. For the most part, we have worked with already implemented architectures during the semester, and we thought it could be interesting to explore the deep learning topic more in detail. Unfortunately, we arrived to the conclusion that this process would take us too much time, as models need to be trained.

After debating, it was decided that the first two options were the most viable ones.

3. First feedback: 5 min presentation

After the brainstorming process, we took advantage of having the opportunity to give a 5-minute presentation about our different proposals, which helped us decide what would be the best option for our activity.

In our presentation, we talked about our goal, that was to teach some new imaging modalities through a competitive and collaborative game. After setting the goal, we presented the two game options to the class:

- **Bingo:** the bingo game consisted of completing a table full of numbers by getting some random numbers after every task was fulfilled. It had the advantage of randomness, which would make the game even more interesting.
- **Word finders:** on the other hand, the word finders game consisted of providing letters to help the teams discover a secret word after each task was completed. This had the advantage of being more “controllable” to us in terms of time, since we wanted to ensure that most teams had time to finish.

After explaining our ideas, and having all this in mind (that it would be a competitive game played by teams, in which the goal would be to learn about different medical imaging modalities), we asked the class what of the two game options would motivate them more or helped them learn the most. Moreover, we asked them to provide us some feedback through a mail, to facilitate our work in order to consider all they had to say.

Unexpectedly, the result was tied, so it was our choice to decide whether if bingo or word finders was the best option. Nevertheless, we also took into account the feedback provided by many of our classmates. In general, they supported both options, but some were worried about the time that each challenge will take, since it was true that there were many modalities and possibilities to work on.

We read the feedback from our classmates and after some discussion, we got to the conclusion that it was possible to combine both ideas. This way, we would make a profit of the randomness of the bingo game, which adds an unpredictable factor that could surely motivate the participants, as well as the “control” that the word finders game gave us. In this case, including the word finders game allowed us to have control over timing and the development of the activity in case some groups needed more help or time. However, this also maintained the random choice of tasks given by the bingo, which was an added fun aspect.

4. Chosen idea

Based on the initial feedback received from the students, we decided to incorporate the concept of a Bingo gameplay challenge alongside a word-guessing activity. The primary objective of our eHealth Lab was to engage participants in a competitive and educational game involving medical imaging tasks. The goal was to enable participants to learn and apply specific methods commonly used in medical imaging domains. By conducting the lab in a game format, we aimed to foster a sense of competitiveness and achievement among the participants.

To ensure the lab met its intended objectives, we established the following requirements:

1. The game should be competitive among different groups.
2. The tasks should involve a variety of medical imaging activities that participants need to complete.
3. Successful completion of each task should reward participants with hints that can be used to progress and ultimately win the game.

4. The game should be time-bound, with the difficulty of each task calibrated to ensure participants can complete them within the allotted time.
5. The game should incorporate both a Bingo card gameplay mechanic and a word-guessing challenge.

The final chosen format for the lab is titled **Word Finders with Bingo Game**. The game begins with each group receiving a Bingo card containing numbered slots. Each number corresponds to a specific medical imaging task that the group must complete. For every Bingo achieved (three tasks completed in a row), the group earns letters as hints. These hints are used to deduce the final mystery word. The group that successfully guesses the mystery word within the given time limit is declared the winner.

5. Planning

Once the lab activity was defined and the game procedures were finalized, we developed a detailed plan for executing the lab. The following tasks were outlined to ensure the successful implementation of the lab:

1. Creating the Bingo Gameplay Website

Initially, we considered creating physical Bingo cards and conducting the game manually. However, after further discussion, we decided to make the lab more interactive by creating a digital Bingo gameplay website. The advantage of this approach was that selecting a number directly on the website could lead participants to the corresponding medical imaging tasks seamlessly. Additional features were incorporated, such as marking completed tasks by turning the associated number green and displaying celebratory visuals when participants achieved a Bingo. Keeping these features in mind, we successfully created the Bingo gameplay website to enhance the overall experience.

2. Developing Google Colaboratory Notebooks for Tasks

For the medical imaging tasks, we deliberated on which domains to include and finalized the following: CT, MRI, X-ray, PET, and Ultrasound. Each member of our team contributed to the creation of the Colaboratory notebooks. The notebooks were designed to include prewritten code snippets as well as unfinished scripts that participants would complete to finish the tasks. A total of nine notebooks corresponding to the Bingo card numbers were created, ensuring a diverse range of challenges within the medical imaging domain.

3. Preparing Datasets for Each Task

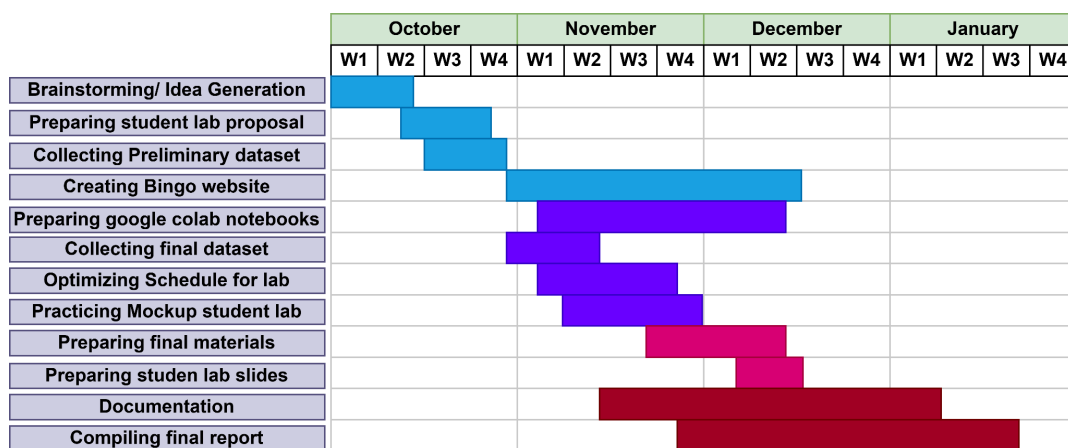
Datasets play a critical role in medical imaging tasks. To ensure quality, we conducted thorough research and literature reviews to identify publicly available datasets that are widely used in the field. These datasets were then incorporated into the notebooks. Finally, we consolidated all the datasets into a single folder, allowing participants to easily access and import them while completing the tasks.

4. Selecting the Mystery Word

As the final task of the game required participants to guess a mystery word, we aimed to select a challenging and domain-relevant term. After brainstorming various deep learning-related words such as Random Forest, ResNet, Weight Initialization, and Cross Entropy, we ultimately chose "**Xavier Initialization**" as the mystery word. This word was selected for its complexity and relevance to deep learning.

The planning phase of the lab activity was focused on creating an engaging, interactive, and educational experience for the participants. By developing a Bingo gameplay website, preparing Colaboratory notebooks with diverse tasks, selecting appropriate datasets, and choosing a challenging mystery word, we ensured that the lab would meet its objectives. The combination of competitive gameplay and task-based learning was designed to foster active participation, collaborative problem-solving, and an enhanced understanding of medical imaging techniques and deep learning concepts.

In order to make our work more efficient, we divided the tasks that we had and put deadlines to finish the work on time. The following figure shows the Gantt chart that we used.



6. Plan execution and lab preparation

○ Notebooks creation

Nine Colab notebooks were developed, corresponding to the nine numbers on the bingo website. Each notebook focused on a specific imaging modality, namely MRI, CT, Ultrasound, PET, and X-Ray. The data for each modality was organized into distinct paths, and shortcuts were provided to users' Colab environments to simplify and expedite the data downloading process.

Each notebook takes approximately 5 minutes to complete, but users are not required to finish all nine. The goal is to achieve bingo, where completing:

- The first three notebooks earns 2 letters.
- Two additional notebooks (for a second bingo) earns 2 more letters, and so on.

Here is a breakdown of nine notebooks by modality:

Ø MRI Notebooks (2 notebooks)

1. MRI DICOM Analysis:

- Focused on analyzing MRI DICOM images.
- Users extracted metadata such as echo time and slice thickness and calculated their multiplication.
- This notebook taught the foundational skills needed for working with MRI DICOM data.

2. Knee MRI Visualization:

- Visualized a knee MRI image and identified rotated slices.
- Combined anatomical understanding with practical skills in visualizing MRI data.

Ø CT Notebooks (2 notebooks)

1. CT DICOM Analysis:

- Introduced DICOM data extraction.
- Users identified details like the manufacturer of the CT scanner, patient age, patient sex, and the body part scanned.
- The goal was to familiarize users with analyzing CT DICOM metadata.

2. Brain CT Visualization:

- Visualized a brain CT image and identified its anatomical plane.
- Based on histogram intensity levels (Hounsfield Units), users determined the ranges corresponding to bone and soft tissue.

Ø PET Notebooks (2 notebooks)

1. PET Intensity Thresholding:

- Focused on lung PET images.
- Users identified intensity thresholds to visualize regions with high metabolic activity.
- The notebook introduced bounding boxes as a tool for analyzing thresholded areas.

2. PET Anatomical Analysis:

- Explored PET slices in different anatomical planes.
- Users identified anatomical regions exhibiting increased metabolic activity.

Ø X-Ray Notebooks (2 notebooks)

1. X-Ray Histogram Equalization:

- Analyzed X-ray images and applied histogram equalization for contrast enhancement.
- Users calculated the average pixel intensity after enhancement.

2. X-Ray Filtering Techniques:

- Applied filters such as median and Gaussian to the original image.
- Users compared the average pixel intensities of the filtered results.

Ø Ultrasound Notebook (1 notebook)

1. Ultrasound Image Processing:

- Analyzed ultrasound images using techniques like histogram visualization, mask application, Gaussian blurring, and edge sharpening.
- Users calculated statistical values such as the mode and median of pixel intensities after processing.

○ Bingo game with interactive website

By combining education, participation, and teamwork into a gamified digital platform, the interactive bingo game was created to accomplish a number of goals. The "Bingo Game with Interactive Website" seamlessly blended traditional bingo gameplay with modern technology, creating a platform for learning, teamwork, and competition. Participants were divided into sub-groups and given virtual bingo cards accessible through the interactive website. Each number on the card was linked to a task, such as solving a question or completing an activity, which participants needed to complete to progress in the game.

As groups completed tasks, they marked off numbers on their bingo cards. Completing a bingo rewarded them with secret letters, which were used to guess a mystery word. The more bingos completed, the more letters a group earned, improving their chances of solving the word. The first team to correctly guess and submit the word was declared the winner.

The use of technology played a central role in enhancing the gameplay. The interactive website provided an intuitive interface where participants could navigate their bingo cards, access tasks, and track their progress. Clicking on a bingo number opened a corresponding notebook containing the task. This system not only modernized the bingo experience but also encouraged participants to utilize digital tools effectively, including accessing datasets through predefined paths.

Teamwork was a key aspect of the game, as groups collaborated to solve tasks and strategize their moves. The time limit of 30 minutes required participants to manage their time efficiently, prioritize tasks, and make quick decisions. This fostered critical thinking, problem-solving, and effective communication among team members.

The competitive nature of the game added excitement, motivating participants to work diligently while enjoying the process. By gamifying education, the bingo game turned learning into a fun and engaging activity. It provided an opportunity for participants to

develop skills in collaboration, strategy, and technical problem-solving while maintaining a sense of enjoyment and accomplishment.

○ Guessing word

One of the most exciting and strategic elements of the "Bingo Game with Interactive Website" was the challenge of guessing the mystery word. The game's progression revolved around completing bingos to earn secret letters, which served as clues to reveal the hidden word.

How It Worked:

1. Completing Bingos for Letters

Each time a group completed a bingo on their virtual card, they were rewarded with two secret letters. These letters were part of the mystery word and were given in specific positions to help participants piece the word together.

2. Collecting More Letters

Groups could complete multiple bingos to gather additional letters, increasing their chances of correctly guessing the mystery word. The more bingos a group completed, the more clues they had to solve the puzzle.

3. Strategic Thinking

Participants had to think critically and collaborate to identify the possible word based on the letters they had collected. They also needed to decide which numbers on the bingo card to target first to maximize their chances of completing bingos quickly.

4. Submitting the Word

Once a group believed they had solved the mystery word, they submitted their final answer by placing a word card on the professor's table. Submissions had to be accurate, as incorrect guesses disqualified groups from further attempts.

5. Winning Criteria

The first group to correctly guess and submit the mystery word won the game. Teams had to be confident in their guess, as they only had one chance to submit their answer.

The Impact of Guessing the Word:

The process of guessing the word added an extra layer of challenge and excitement to the game. It required participants to:

- Analyze the letters strategically.
- Collaborate effectively to pool ideas and eliminate incorrect possibilities.
- Think critically under time pressure.

This element not only made the game more competitive but also encouraged participants to engage deeply with the tasks, fostering problem-solving and strategic thinking skills.

7. Lab execution

The class was divided into groups based on their respective e-health teams. Additionally, lab assistants from the Word Finder Lab were assigned to each group to provide guidance and support during the activity.

Preparation:

On the day of the lab, a brief 5-minute presentation was conducted to explain the structure and flow of the activity. During this session:

- The objectives of the lab were outlined.
- Instructions for the activity were provided.
- Word cards were distributed to each group.

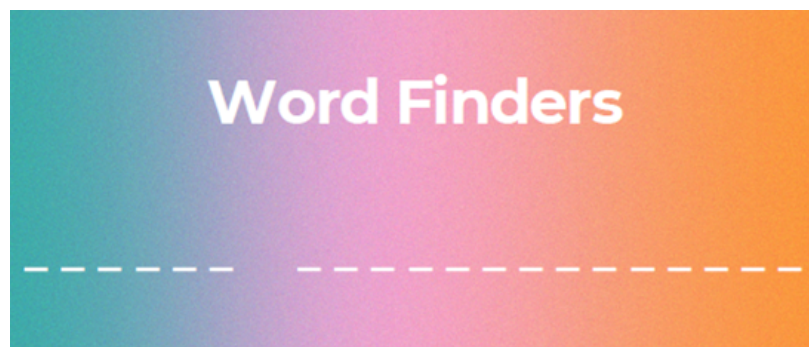


Figure 1. Word card

Activity Execution:

1. **Website Access:** Each group was directed to locate a specific website link on the Moodle platform.

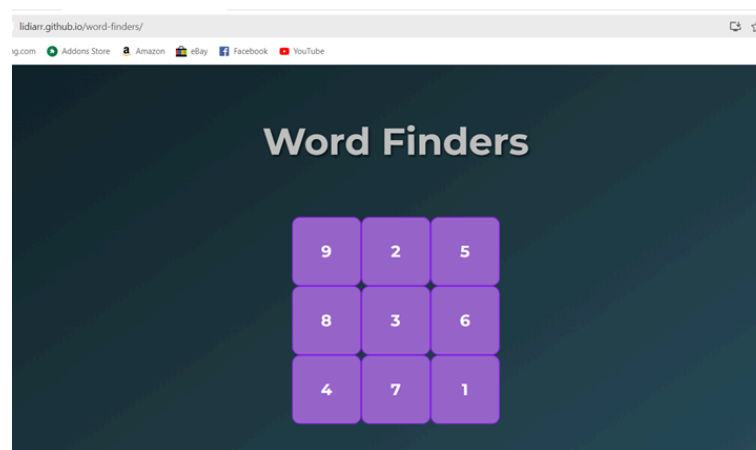


Figure 2. Website

2. **Word Bingo Game:** The activity consisted of multiple rounds of a word bingo game. In each round, specific letters were revealed to help groups deduce the final word. The sequence of letters provided was:
 - Bingo 1: **A, O**
 - Bingo 2: **E, Z**
 - Bingo 3: **V, N**
 - Bingo 4: **R, T**
3. **Final Word Discovery:** By combining the letters revealed across all rounds, the groups were expected to deduce the final word, which was **“XAVIER INITIALIZATION.”**

Completion and Winner Announcement:

Once a group identified the final word, they were required to present their completed word card to the professor at the designated table. In the last 5 minutes of the session, the winner of the activity was announced, and a prize was awarded to the victorious group.

This structure ensured an engaging, interactive, and collaborative learning experience for all participants.

8. Second Feedback

To get an idea of how the participants perceived the lab, we created a google form to obtain feedback about our lab from them. Only 4 people participated in the survey, as the feedback was not mandatory. All the participants found the Bingo Word Finders Game interesting, dynamic and interactive. Similarly, they thought that the well-prepared notebooks interactive with bingo and guessing the word competitively combined fun with the learning process.

Some feedback we received are presented here:

Very interesting and dynamic

Very interactive lab, good job on making solving the code interactive with both bingo and guessing the word.

Interesting game. Choice of the word to be guessed was difficult. So the overall game was fun.

I really enjoyed your lab..the games were so nice, the notebooks are very well prepared..and it was competitive ..thank you ❤️ ❤️

Figure 3. Feedback form answers

9. Teamwork

Teamwork is a key skill for any career and usually plays a central role in projects. In this activity, we focused on our ability to work as a team to solve challenges and provide a good

quality result. We all participated in the decisions to be made, distributed the workload equally and aided in fusing all the work done to form the final result. This allowed us to combine our separate strengths and to learn from our teammates in a learning environment.

To begin with, during the brainstorming phase we all contributed to the ideas proposed. For each of the proposals, one group member presented it and the rest gave their opinion and tried to look at the positive and negative points. In addition, key questions were asked to let the person presenting the idea expand on those points and discuss with the group what the best approach would be. For example, when proposing the idea of bingo, as a group we discussed what the challenges for each number could be and we decided that using what had already been mentioned about different types of image modalities would be interesting.

Once we had to decide on a final idea, between all of us we discussed the final details and we proceeded to distribute the workload. We each got assigned an image modality, as there were 5 of them that we wanted to study and there were 5 group members. For each modality, two tasks had to be made, with the exception of Ultrasound, as the group member assigned to that modality was in charge of making the interactive website.

While working on our respective tasks, we asked each other for advice, specially in some cases where the tasks had elements in common. For example, both in CT and MRI the tasks used DICOM files at some point and used a slider to visualize the image. The two group members working on those tasks were able to comment on the implementation and save time. Additionally, when all the notebooks were finished, we standardized them so it would be less confusing for our classmates. We then structured our drive folders with the notebooks and datasets the same way and we checked our teammates files were also in order.

Another step at this point was to make sure the interactive website worked well and that all the data was right. All the team members contributed to this phase, providing feedback and finding various bugs. Some of the feedback given included some kind of confetti animation when a bingo is made and the possibility to allow multiple answers for a single bingo question. On the other hand, bugs found by various group members included data not being saved if the user reloaded the page, links to notebooks that were repeated while some were missing and answers that had an incorrect value. This showed how it is important for everyone to help each step, as one person can often miss mistakes made. This allowed the member working on the website to develop a good quality website.

Another thing we had to consider during our lab preparation was that the individual challenges were not too complicated nor long, therefore each group member got assigned two notebooks they hadn't seen before and we completed them to ensure they were doable in about 5 minutes or less. As the final preparation before the activity, we all made sure to understand the challenges that the other group members had written and in case of doubt, we talked amongst ourselves to discuss confusing details.

Finally, on the Student Lab day, each teammate got assigned 1 or 2 groups of students to help. However, we kept constant communication between us to know how all the students were doing, to ask each other some notebook specific questions and to make some last

minute changes like extending the timer 5 minutes. This communication was key in order to provide our classmates with a great activity.

10. Conclusion

The Student Lab activity was successfully performed, as participants learned about image processing for different medical image modalities while having a fun time. The game provided participants with a challenge and motivated them to complete as many activities as possible. This allowed them to work with image modalities they had not worked with before, such as X-Ray, PET and Ultrasound.

The structure of the Lab proved to be well structured. The website allowed each team to work independently while having all the information they needed for the game in one place. Having a person assigned to each group also offered them guidance when they had any problem and we were sure everyone was learning and having a good time. However, for the most part, groups did not need much help, as we had prepared the notebooks carefully in order to provide a detailed explanation of each task. Finally, the secret word was complex enough that the teams had a hard time guessing it, which forced them to keep working on the notebooks. However, it was not too complicated as 4 different groups were able to find it.

The feedback from our classmates confirmed the success of our activity and showed us that the effort we put as a team gave a good final result. We achieved our initial goals of making an educational game that would teach our friends about something new in an entertaining and original way.

In conclusion, this activity provided an opportunity for participants to learn more about MRI, CT, PET, X-Ray and US through some fun challenges, while competing with others in a fun environment. There was a balance between education and entertainment that allowed this, and the work of our team made this possible.

11. References

- [1] *Blender* [Computer Software]. <https://www.blender.org/>
- [2] CT Data: [CT Medical Images](#)
- [3] R. M. Summers, "NIH Chest X-ray Dataset of 14 Common Thorax Disease Categories," 2019.

12. Annexes



Project team



Photos in the day of lab